

Where a quantum or generative advantage can, and cannot, live

The structure that makes SQD classically *verifiable* is the structure that makes it classically *constructible*.
Advantage survives only where that equivalence breaks.

Classically verifiable & simulable

CLASSICAL WINS

Task: molecular ground-state SQD

- SHCI / CIPSI / DMRG / AFQMC reach the ~1.3 mHa consensus at lower cost
- LUCJ circuits dequantized on a laptop; random+recovery matches
- exponential even for a perfect sampler in delocalized states

The generative advantage is dequantized along with the sampler.

Classical signal degrades

OPEN / CONDITIONAL

Where the cheap prior fails:

- high noise → starved valid-shot budget; symmetry-valid generators waste nothing
- strong multireference → HF-based perturbative ranking misfires
- amortization across a PES / chemical series (train once)

The empirical target of \$6–\$7 (GFlowNet).

Provably hard classically

ADVANTAGE = THEOREM

Task: learning from experiments

- two-copy / quantum-memory learning: exponential separation, **unconditional** (a theorem, not a conjecture)
- cannot be dequantized

Caveats: tasks are contrived;
fragile on NISQ hardware;
useful reach = device characterization

The highest-value open bridge to chemistry.

cheap classical construction & verification

classical construction provably expensive

→ aim generative & quantum methods where classical proof is NOT cheap →